Benign epithelial tissue neoplasms

1- Squamous cell papilloma
2- Keratoacanthoma

1.Squamous cell papilloma



➢ **Definition**

A benign neoplsm of stratified squamous epithelium.



➢ **Etiology**

Human papilloma virus

Type 6,11

1.Squamous cell papilloma

➢ Clinical picture

Age: at any age.

Site: any oral site as Lip, tongue, palate.



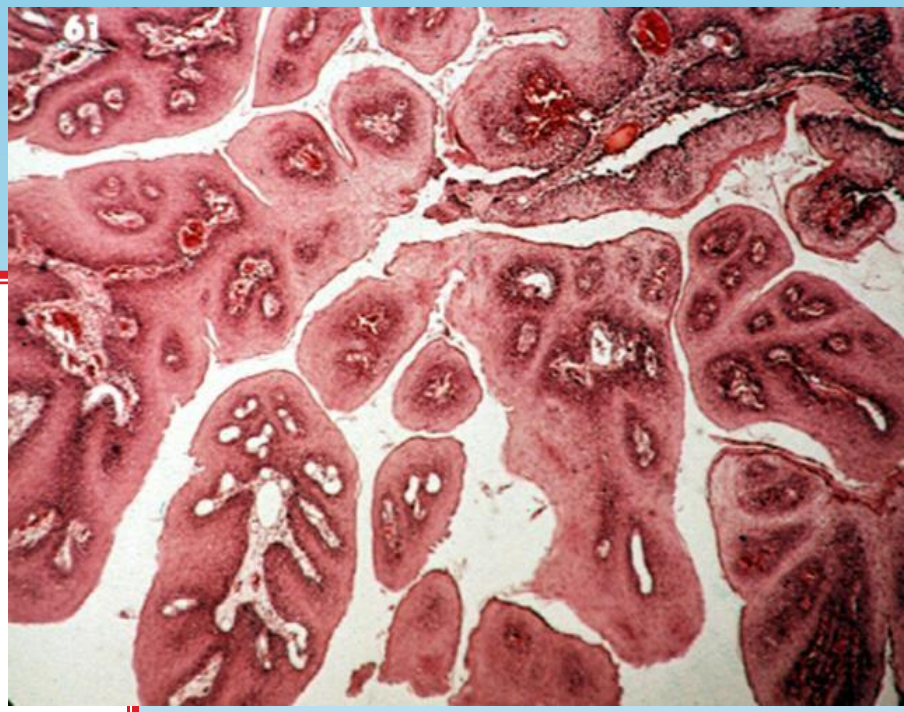
Shape: exophytic mass, sessile or pedunculated Forming finger like projections (verrucus or "cauliflowerlike"appearance).

Color: white, or normal in color depending on the amount of surface kertinization.

Consistency: soft.

Symptoms: asymptomatic.

1.Squamous cell papilloma



➢ Histopathology

- It consists of hyperplastic stratified squamous epithelium forming fingers like projections.
- The epithelium may show hyperkeratosis.
- The epithelium covers a thin core of connective tissue.
- The CT shows chronic inflammatory cells.
- Few koilocytes



2.Keratoacanthoma (self healing carcinoma)

➢ **Definition**

It is a benign neoplasm of stratified squamous epithelium that have strong clinical and histopathologic similarity to well –differentiated squamous cell carcinoma.



2.Keratoacanthoma (self healing carcinoma)

➢ Etiology

The exact etiology is unknown but may be:

- **Human papilloma virus.**
- **Sun rays.**
- **Genetic factor.**

2.Keratoacanthoma (self healing carcinoma)

➢Clinical picture

Age: 50 to 70 years.

Sex: Males more than Females.

Site: Sun exposed areas as face, hands, lips.

It occurs very rarely on oral mucous membrane.

Shape: It is small dome shaped nodule with central depression filled with keratin (keratin plug).

Consistency: firm.

It has a characteristic course:

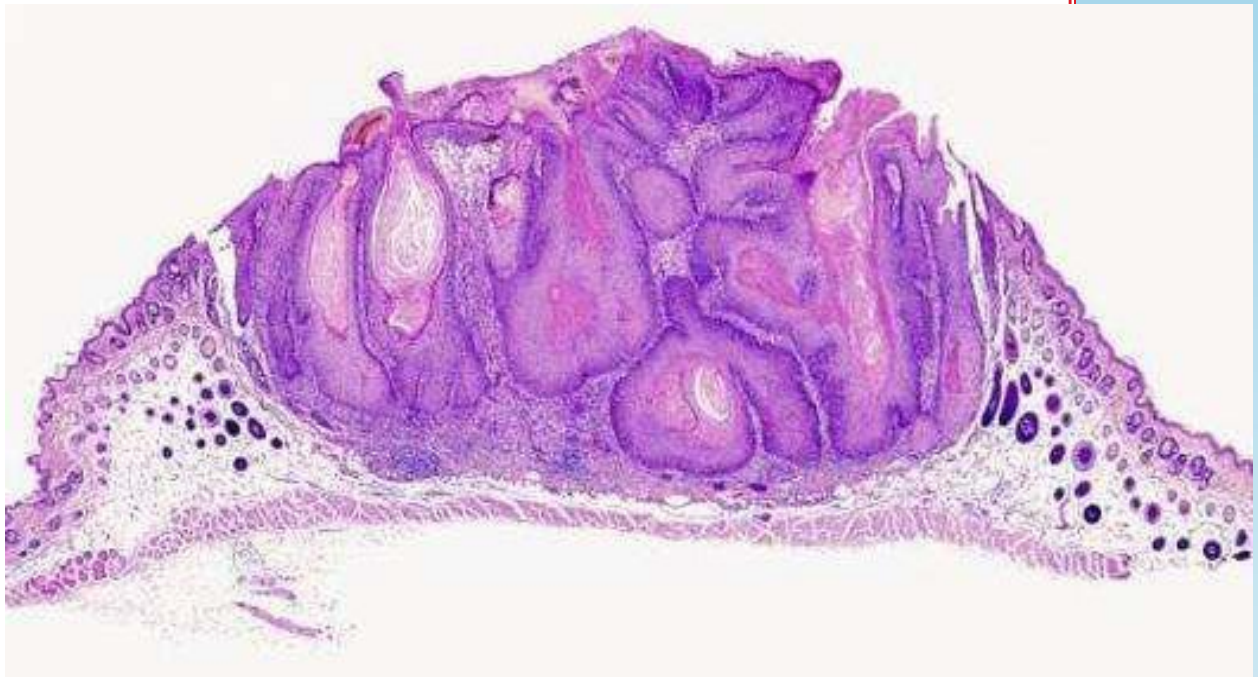
- takes 4 to 8 weeks to reach maximum size (1-2 cm in diameter) .
- Stays 4 to 8 weeks unchanged.
- Spontaneous regression in 6 to 8 weeks (leaving a depressed scar).



2.Keratoacanthoma (self healing carcinoma)

➢ Histopathology

- Hyperplastic stratified squamous epithelium with a central crater filled with keratin .



- The epithelium at the base of the crater proliferate downward which Stimulate chronic inflammatory cells In CT.

- Hyperkeratosis.
- Deep in the central core of the lesion, there is abnormal or premature keratin production like those found in well differentiated squamous cell carcinoma (individual cell keratinization , keratin pearls).

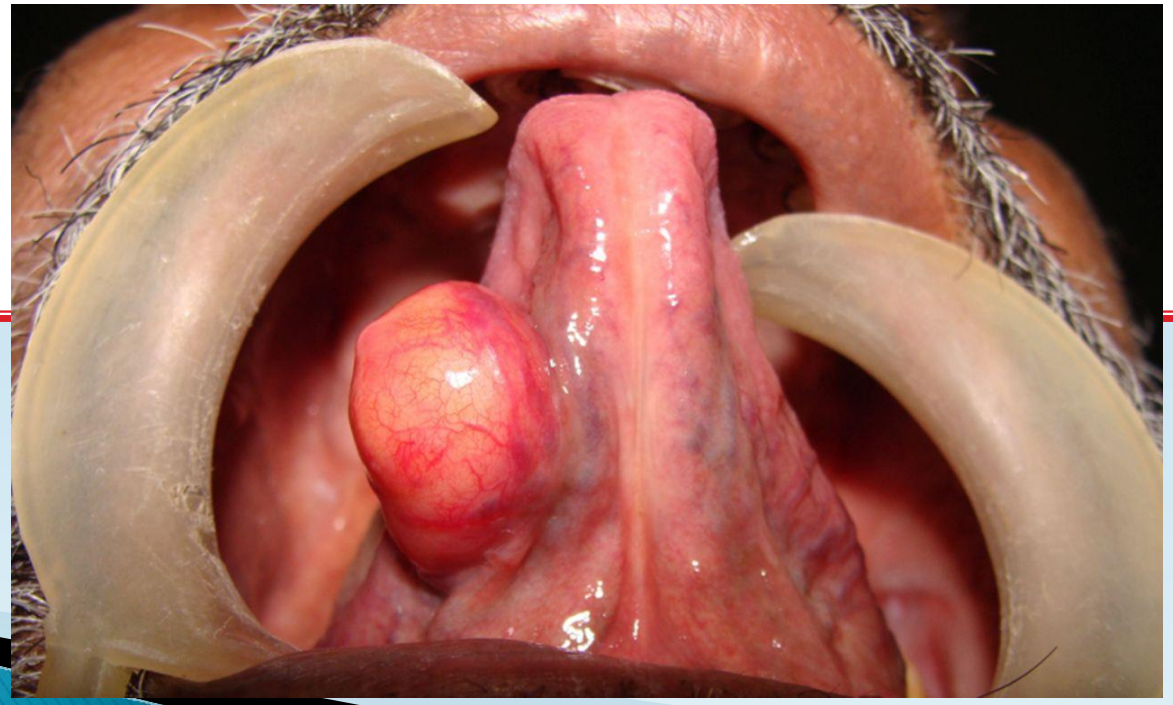
Benign connective tissue neoplasms

1. Lipoma
2. Myoma
3. Neurofibroma
4. Neurilemoma (Schwannoma)
5. Granular cell tumor
6. Congenital epulis of newborn
7. Chondroma
8. Osteoma

1. Lipoma

➢ Definition

- It is a **benign** neoplasm of **fat** (adipose tissue).



1. Lipoma

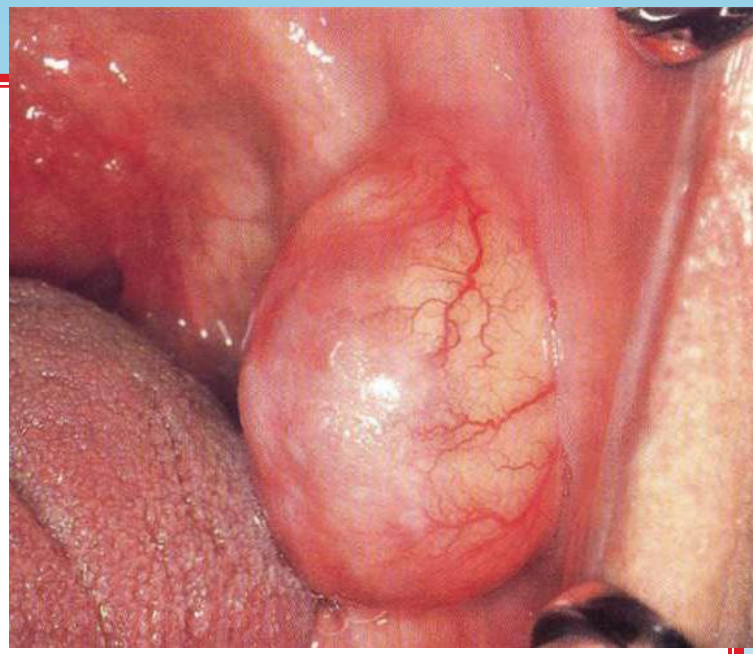
➢ Clinical picture

- **Site:** the most common site
  is the the trunk and extremities.

Intraorally, it may occur on the
tongue, buccal mucosa, buccal vestibule and floor of the mouth.

Shape: soft smooth surfaced mass (nodule).

Color: Yellowish in color.





1. Lipoma

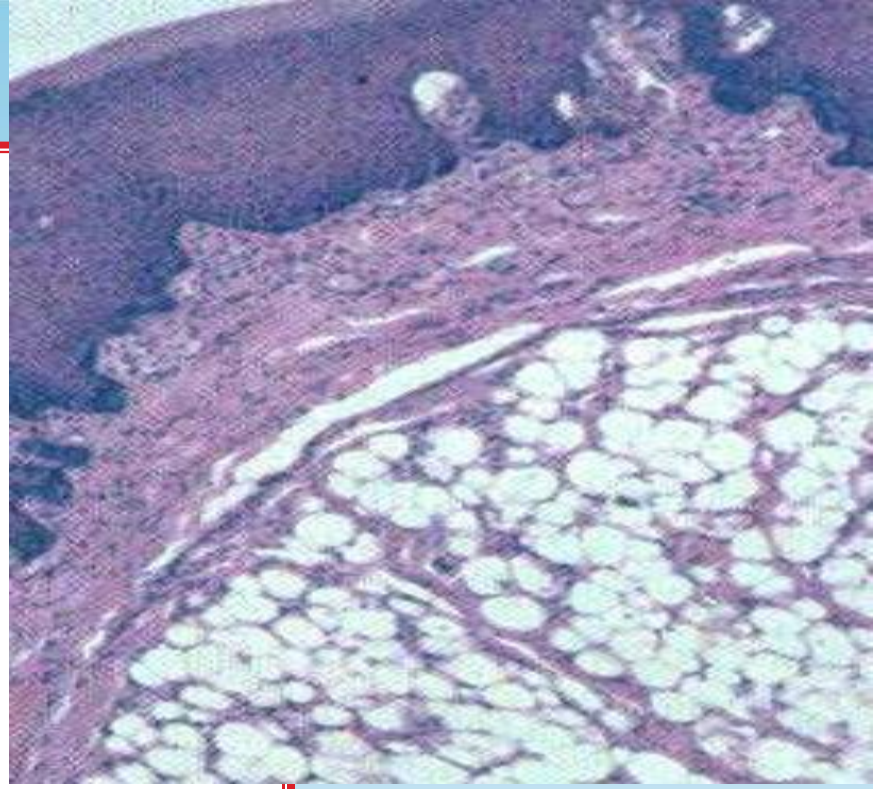
➢Histopathology

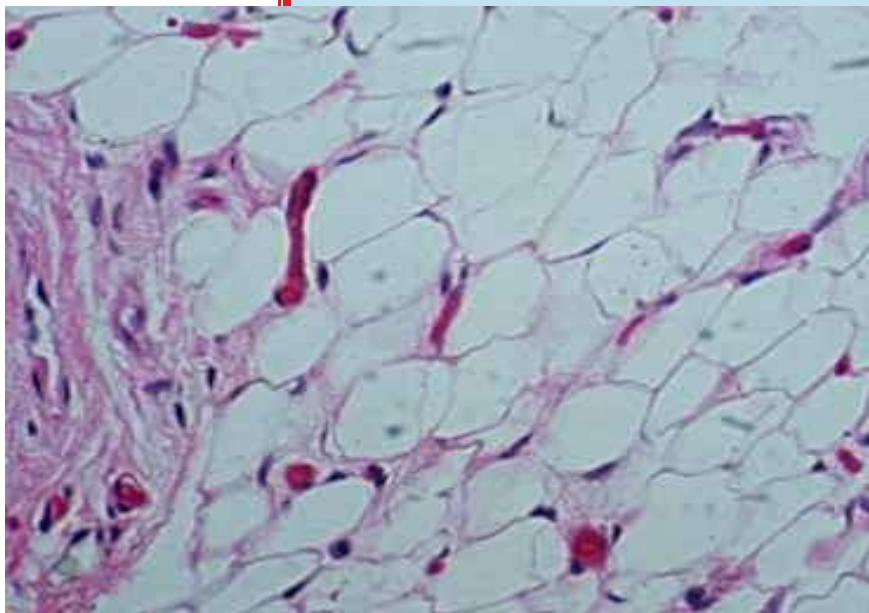
Connective tissue lesion that is formed of:

1. **Fat cells which are round or polyhedral with clear cytoplasm. The nuclei of the cells are compressed against one side of the cell membrane.**

2. **There is a fibrous connective tissue capsule.**

3. **The tumor cells are separated by connective tissue septa extending from the capsule (lobulated mass).**





2. Myoma

➢ Definition

- It is a **benign** neoplasm of **muscle**.

➢ Types

Leiomyoma: benign neoplasm of smooth muscle.

Rhabdomyoma: benign neoplasm of striated muscle.

2. Myoma

➢ Clinical picture

- Myomas of the oral cavity are rare.

- Most common intraoral site is the tongue.

- Smooth muscle fibers are found intraorally in the wall of blood vessels of the tongue.

- Myomas appear as firm mucosal nodule (mass).



2. Myoma

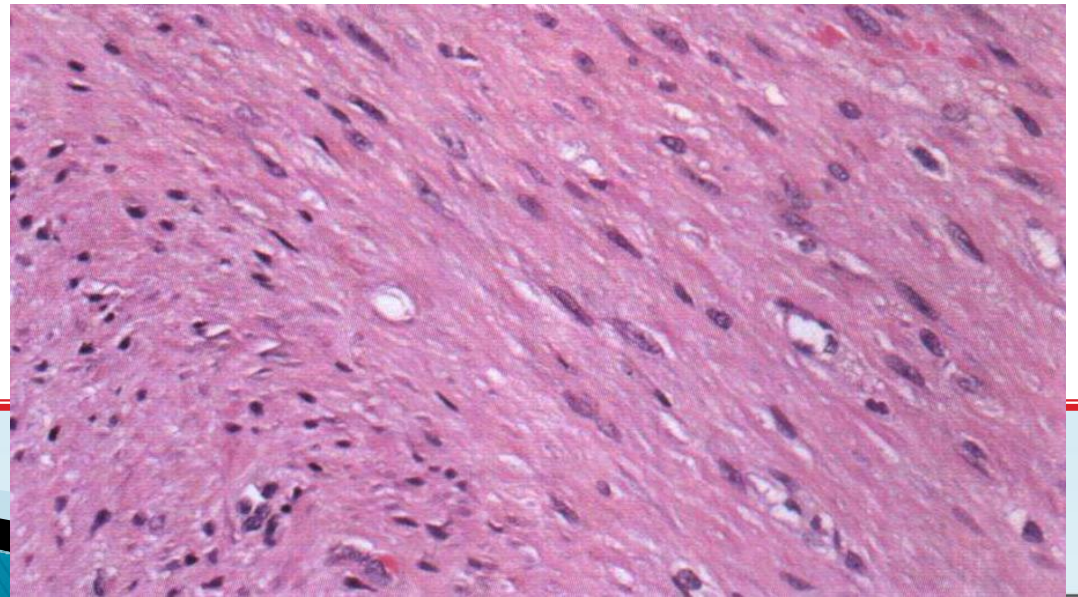
➢ **Histopathology**

Leiomyoma:

Connective tissue lesion that is formed of:

- **Interlacing bundles of spindle shaped smooth muscle cells.**

- **The nuclei are elongated and blunt ended.**



2. Myoma

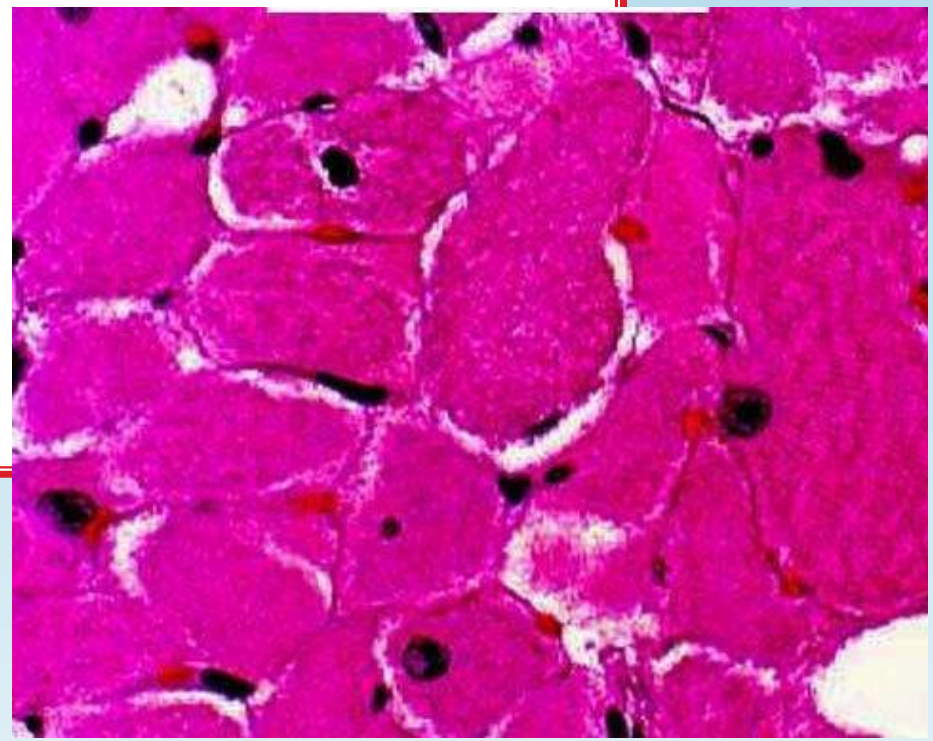
➢ **Histopathology**

Rhabdomyoma:

Connective tissue lesion that is formed of:

- **Large polygonal cells with granular, eosinophilic cytoplasm. with frequent cross striations.**

- **Nuclei are small and round and located both peripherally and centrally.**



3.Neurofibroma

➢ Definition

- It is a benign neoplasm of nerve tissue.

- It arises from a mixture of cell types including Schwann cells and perineural fibroblasts.



3.Neurofibrom

➢ Clinical picture

- Neurofibromas can arise as solitary tumors or multiple neurofibromas (which is a component of neurofibromatosis syndrome)

- **Site:** the most common site is the the skin. Intraorally, it may occur on the tongue and buccal mucosa.

Shape: soft smooth surfaced mass (nodule).



3.Neurofibroma



➢Clinical picture

Neurofibromatosis

(Von Recklinghausen's disease of skin)

- **Patients have multiple neurofibromas that can occur anywhere in the body but are most common on the skin.**

- **Another highly characteristic feature is the presence of "cafe au lait" pigmentation of the skin (smooth-edged, brown macules).**



- **These multiple neurofibromas has a great tendency for malignant transformation.**

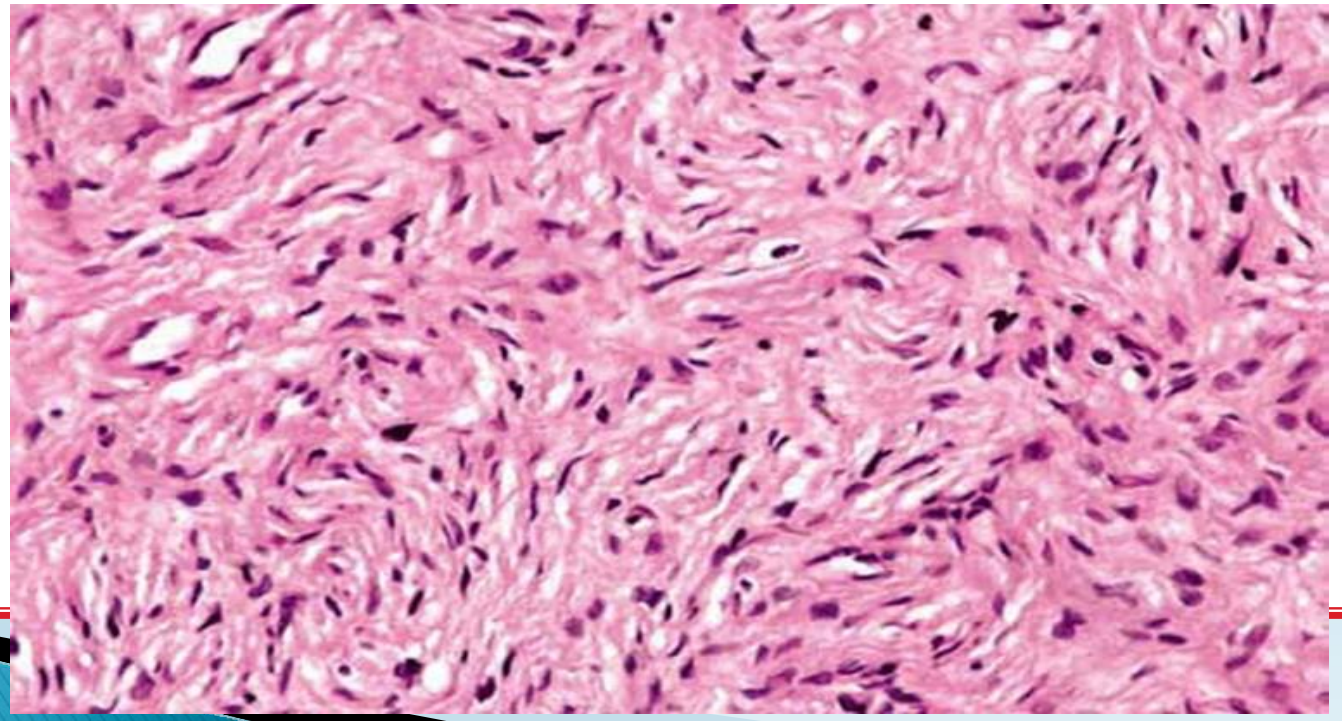
3.Neurofibrom

➢ Histopathology

Connective tissue lesion that is formed of:

1. Proliferating spindle-shaped cells that have wavy nuclei.

2. These cells are surrounded with delicate collagen bundles.

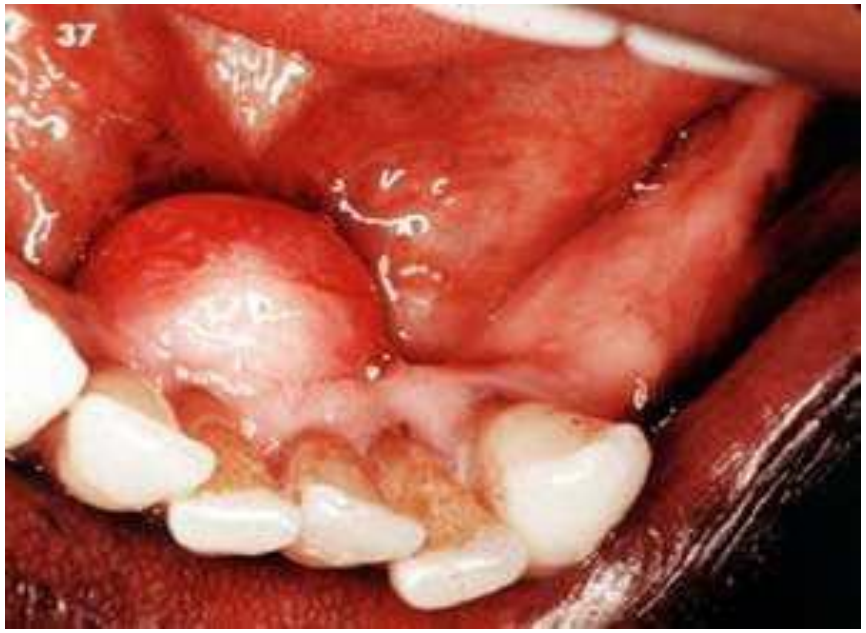


4. Neurilemoma (Schwannoma)

➢ Definition

- It is a benign neoplasm of nerve tissue.

- It arises from Schwann cell of the nerve sheath.



4. Neurilemoma (Schwannoma)

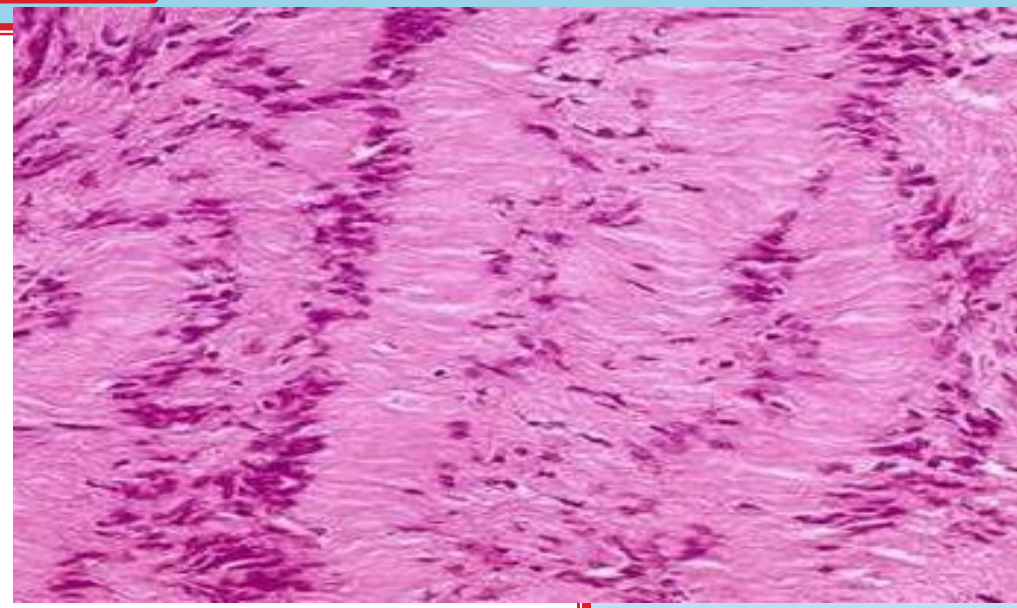
➢ Clinical picture

- **Site:** the most common **Intraoral** site is the **tongue.**

Shape: it arises as a **mass (nodule).**



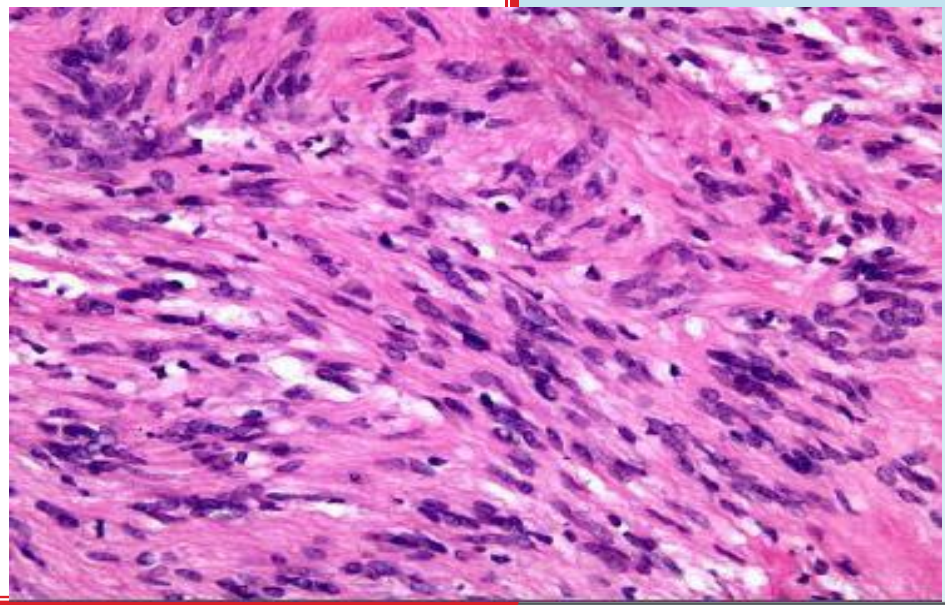
4. Neurilemoma (Schwannoma)



➢ Histopathology

The tumor has two microscopic types:
Antoni A and Antoni B.

1. Antoni A type is characterized by:
bundles of spindle-shaped Schwann cells.
These cells often form a palisaded arrangement around central acellular eosinophilic areas known as Verocay bodies.



2. Antoni B the spindle cells are randomly (haphazardly) arranged.

5. Granular cell tumor

➢ Definition

- It is a benign neoplasm formed of granular cells.

- It was believed to be of striated muscle origin.

- However, recent investigations revealed a nerve tissue origin (derived from Schwann cells)

The granular cells of the tumors
are positive to S-100 protein
that is present in nerve cells.



5. Granular cell tumor

➢ Clinical picture

- **Site:** the most common **Intraoral** site is the **tongue.**

Shape: it arises as a **mass (nodule).**

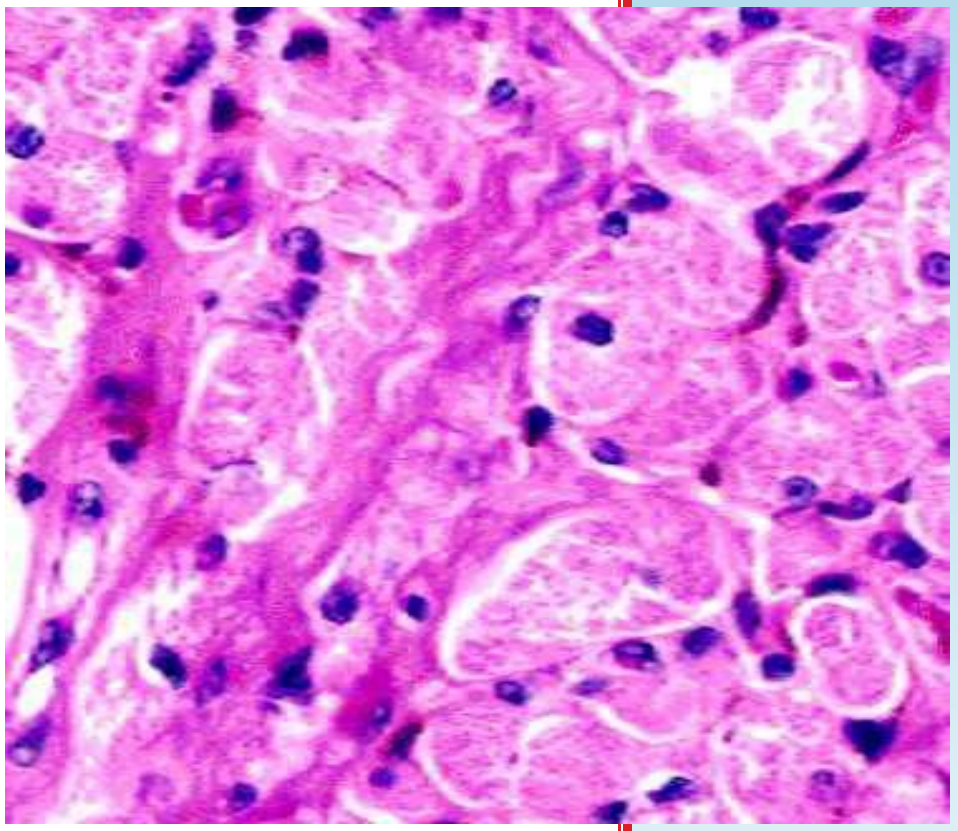


5. Granular cell tumor

➢ Histopathology

Connective tissue lesion that is formed of:

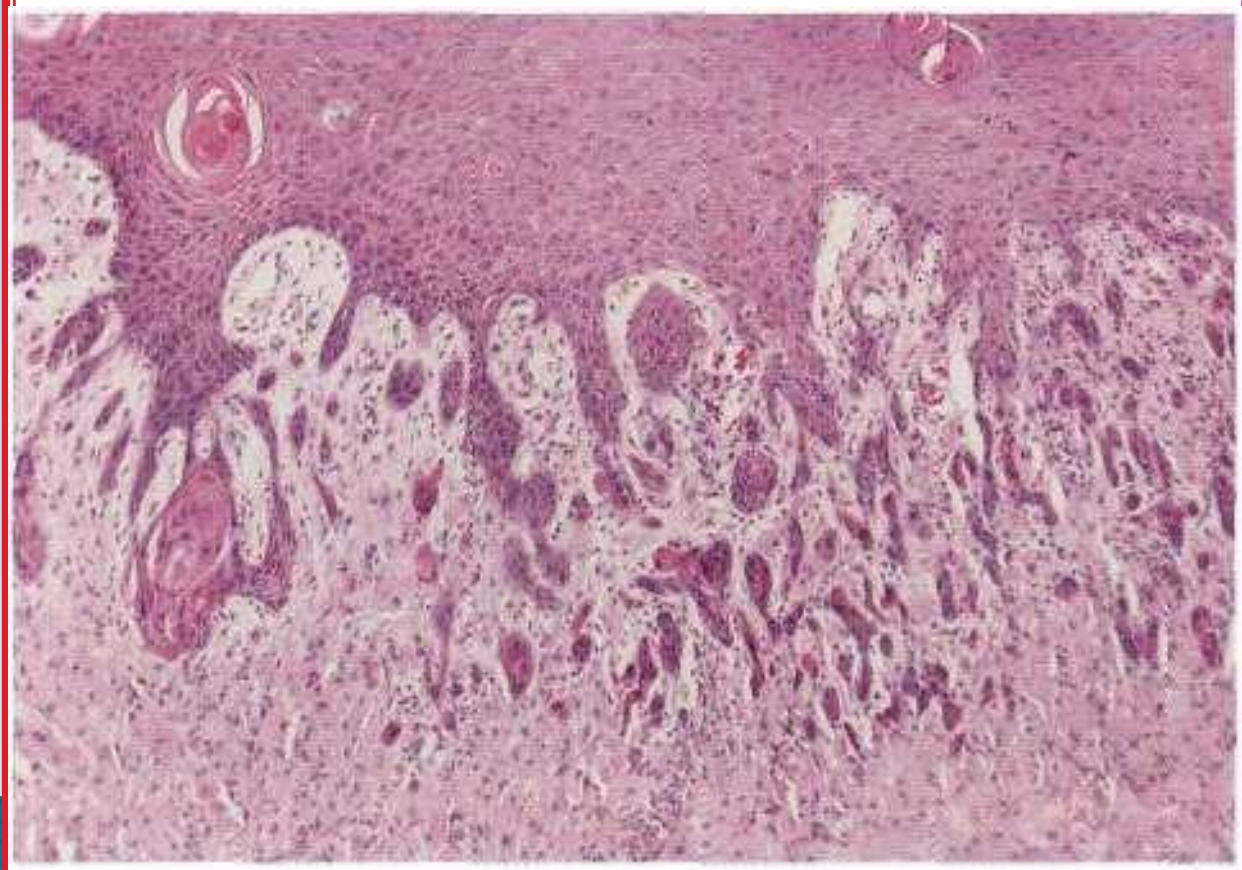
- Sheets or nests of large polygonal cells with eosinophilic, granular cytoplasm and small centrally or peripherally placed nucleus.

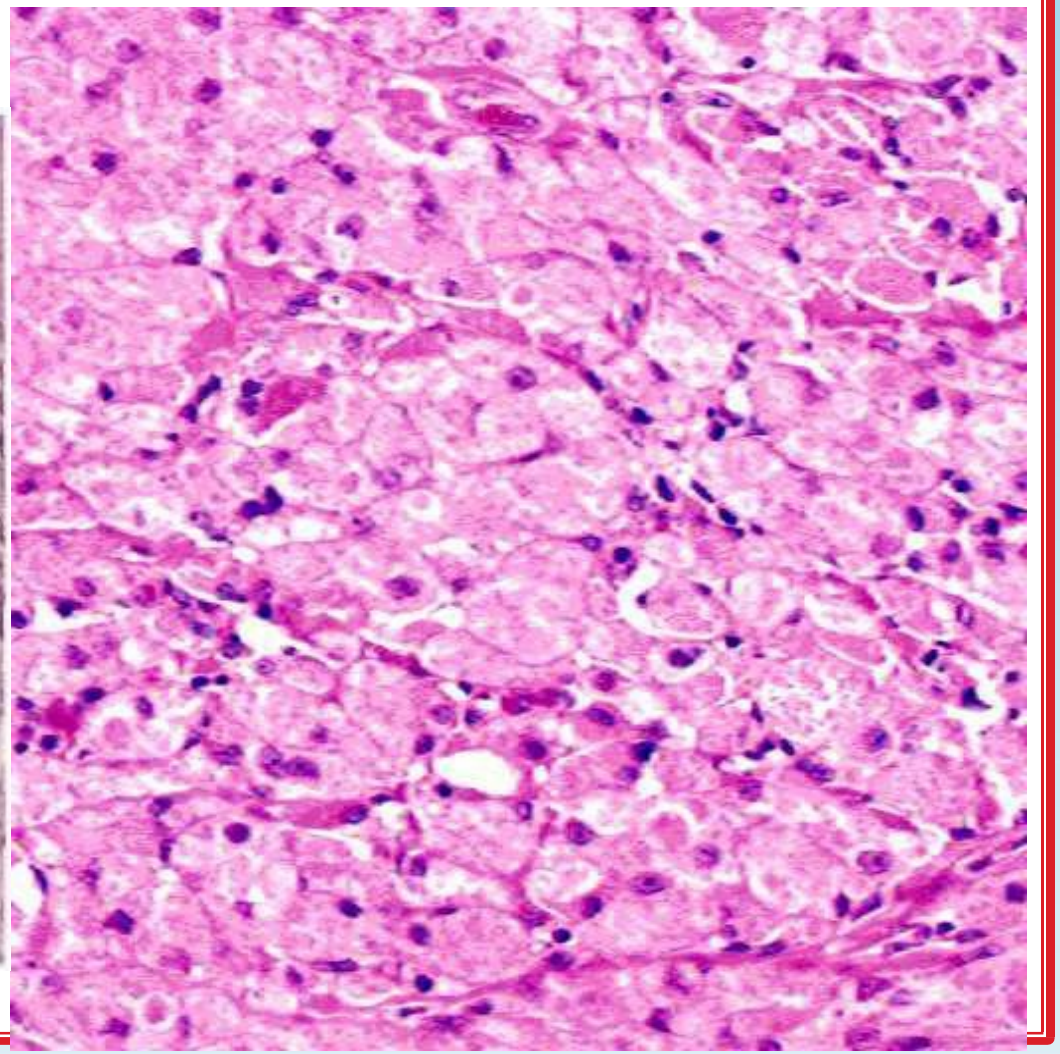


- The overlying epithelium is hyperplastic and may results in a mistaken diagnosis of squamous cell carcinoma (pseudoepitheliomatous "pseudocarcinomatous" hyperplasia).

5. Granular cell tumor

- **Histopathology**





6. Congenital epulis of newborn

➢ Definition

- A congenital tumor that occurs almost exclusively on the alveolar ridges of newborns.

- The histogenesis (origin) of this tumor is unknown.

- The tumor cells are microscopically identical to those of granular cell tumors but:
- The cells of the tumor are negative for S-100 protein (nerve marker) and actin (muscle marker).

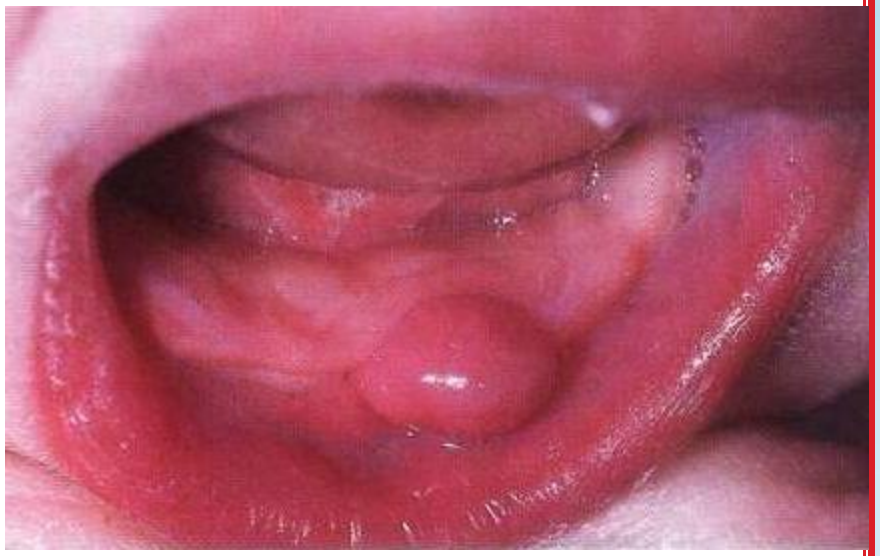


6. Congenital epulis of newborn

➢ Clinical picture

- Appears as a pink-to-red smooth-surfaced mass on the alveolar ridge of a newborn.

- It occurs lateral to the midline and is more common on the maxillary ridge.



- It affects female more than males.

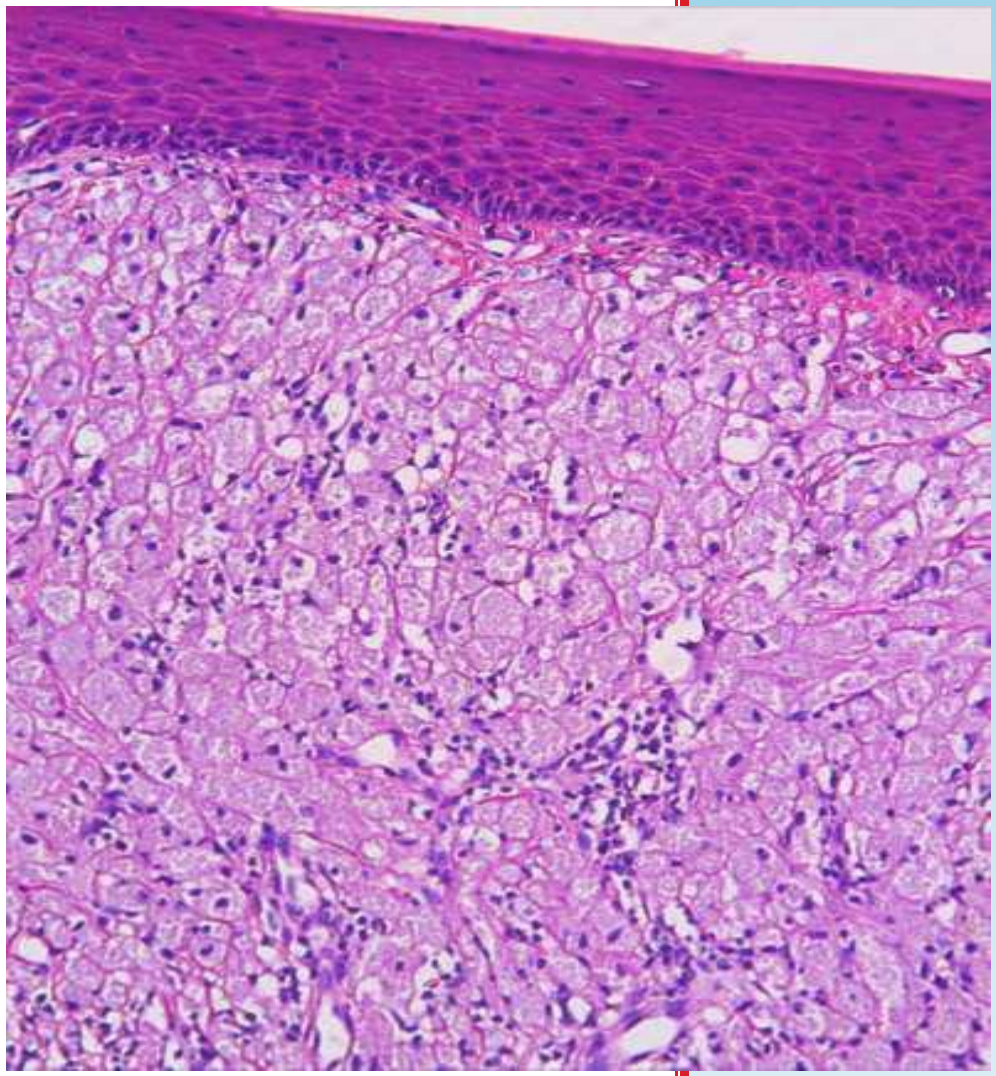
- After birth, the tumor appears to stop growing and may even diminish in size.

6. Congenital epulis of newborn

➢ **Histopathology** Connective tissue lesion that is formed of:

1. Sheets or nests of large rounded cells with eosinophilic, granular cytoplasm.

2. The overlying epithelium never shows pseudoepitheliomatous hyperplasia but have atrophy of the rete ridges.



7. Chondroma

➢ Definition

- It is a benign neoplasm composed of cartilage.
- Chondroma is one of the common bone tumors.

- Gnathic chondromas arise from remnant of cartilaginous rests.
- Such rests are located in the anterior maxilla (in the region of premaxilla), symphysis, coronoid process and condyle.



7. Chondroma

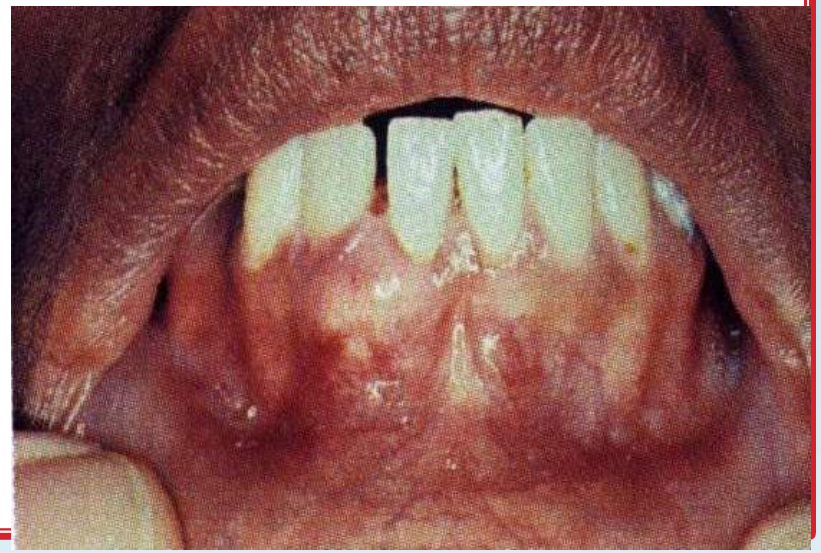
➢ Clinical picture

- **Site:** Gnathic examples have been found in the condyle, coronoid process, symphysis area or anterior maxilla.

- **Age:** It affects adult patients.

- **Consistency:** It appears as slowly growing hard swelling.

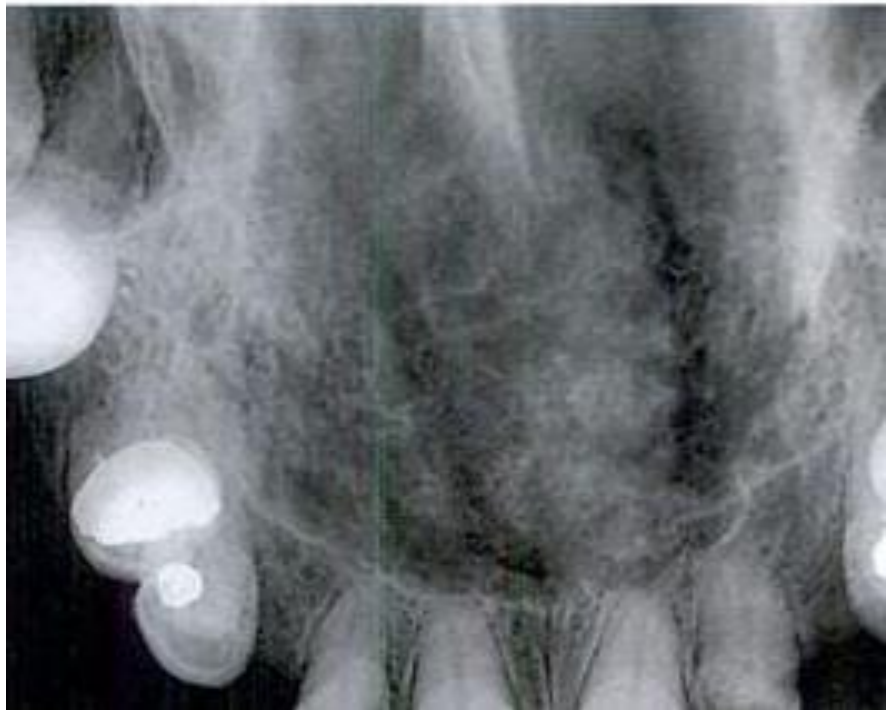
- **Complication:** The lesion may cause loosening of the teeth due to root resorption.



7. Chondroma

➢ Radiographic features

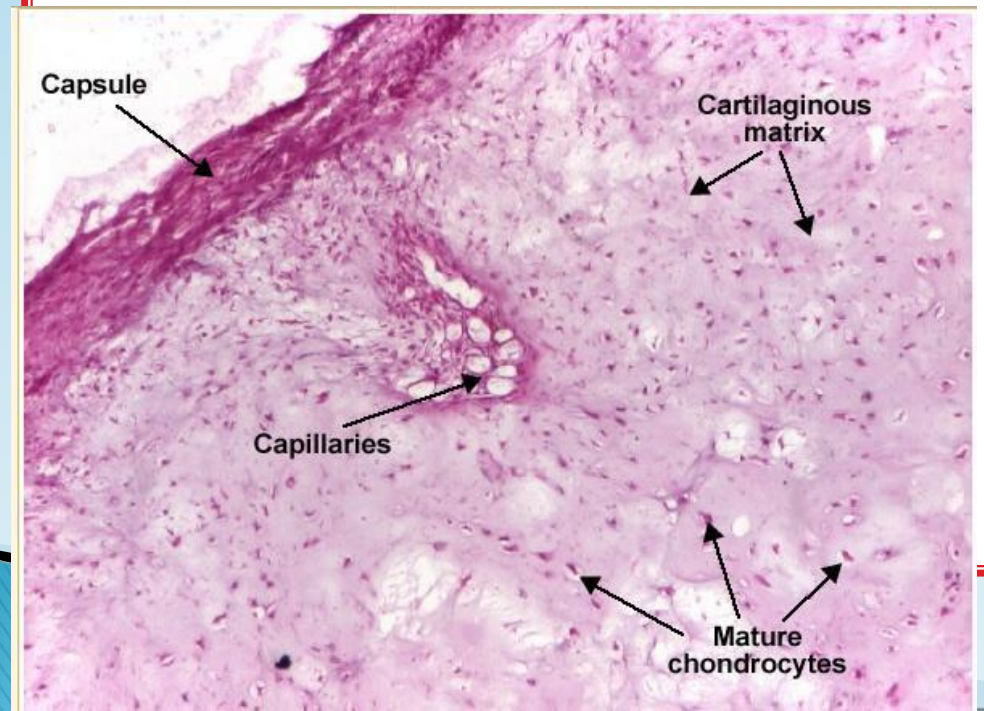
It appears as radiolucency with central areas of radiopacity.

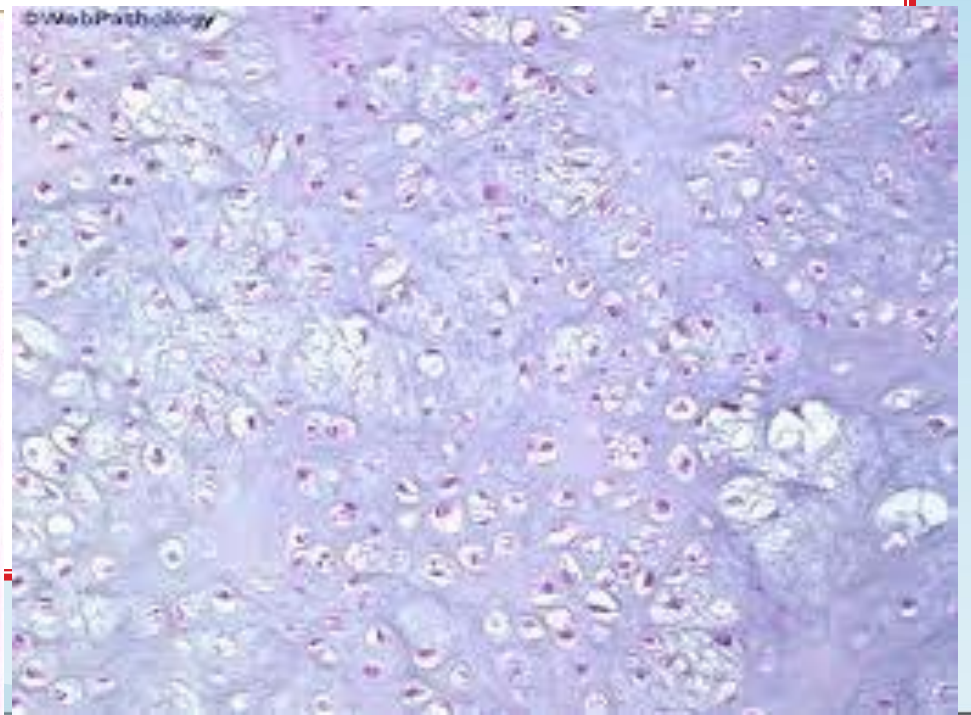


7. Chondroma

➢ Histopathology

Chondroma appears a s a well circumscribed mass of hyaline cartilage (homogenous, clear, basophilic material) that shows lacunae containing small chondrocytes with pale cytoplasm and small round nuclei.





8. Osteoma

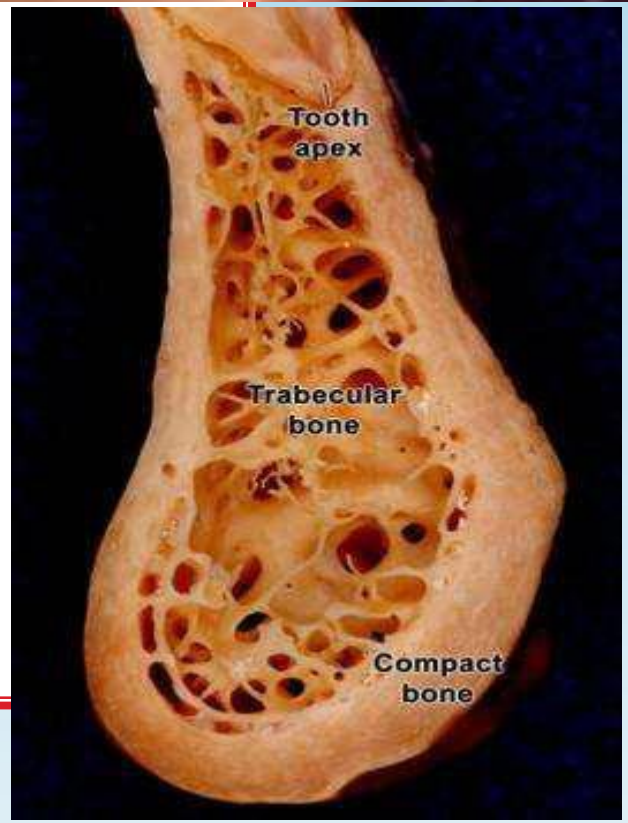


➢ Definition

- It is a **benign** neoplasm of **bone.**

- Tumors composed of mature **compact or cancellous** bone.

- It may be **solitary** or **multiple** (as in case of Gardner syndrome).



8. Osteoma

➢Clinical picture

Osteomas of the jaws may:

1.Arise on the surface of the bone, (periosteal osteoma), And appear as slowly growing hard mass.

or

2. May be located in the medullary bone (endosteal osteoma).

Small endosteal osteomas are asymptomatic, but large lesions cause a slowly progressive enlargement of the affected area.



8. Osteoma

➢ Radiographic features

It appears as a well circumscribed radiopaque mass.



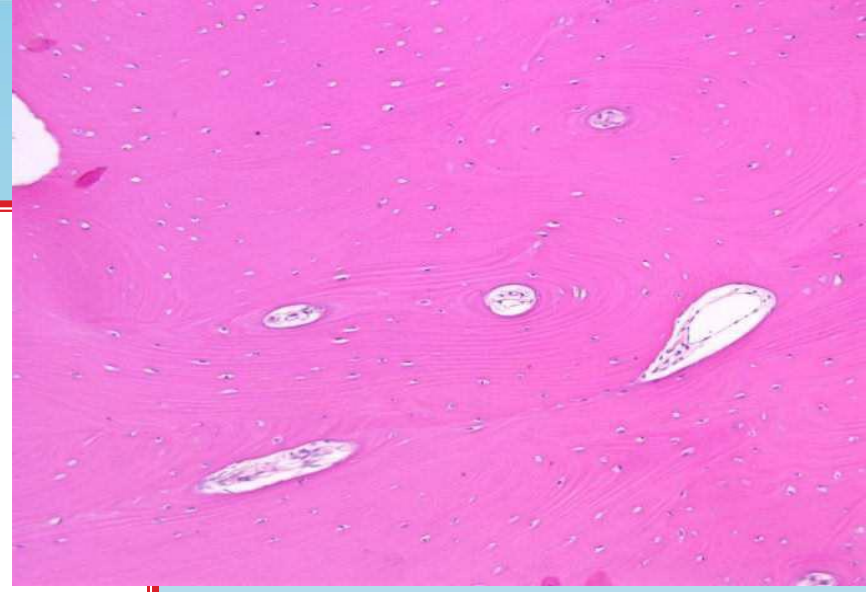


8. Osteoma

➢ Histopathology

- **Compact osteomas** are composed of **dense bone** showing minimal marrow tissue.

- **Cancellous osteomas** are composed of **trabeculae of bone** and marrow tissue.





Hamartomas

- **Harartoma:**

It is a developmental tumor like malformation characterized by presence of normal tissue in normal site, but in excessive amount.

- **Nevus:** it is a hamartoma occurs on the skin or the mucous membrane.

- **Types of Nevi:**

Keratotic Nevus: White sponge nevus.
Pigmented or melanocytic nevus.
Vascular Nevi: Hemangioma and lymphangioma.